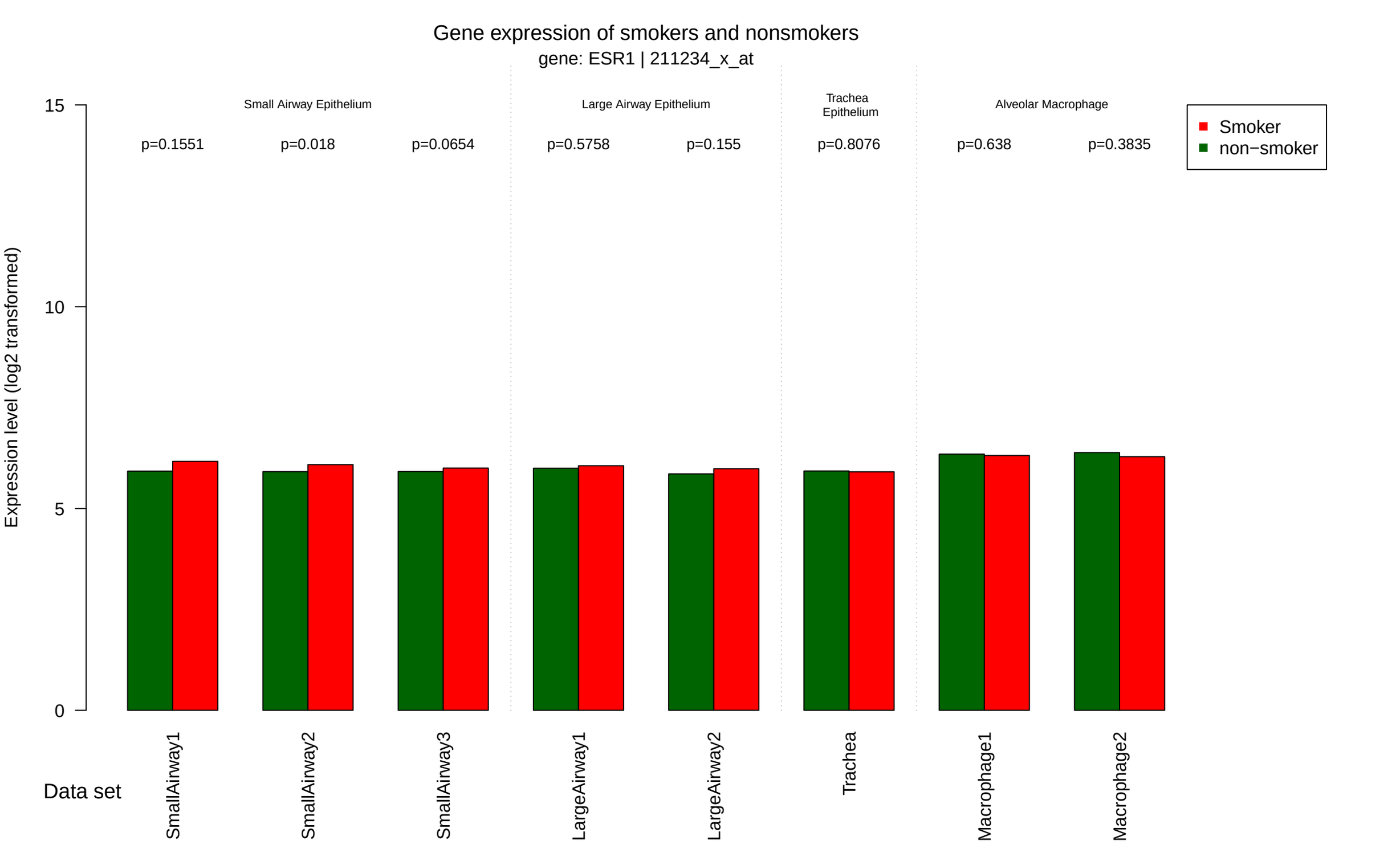
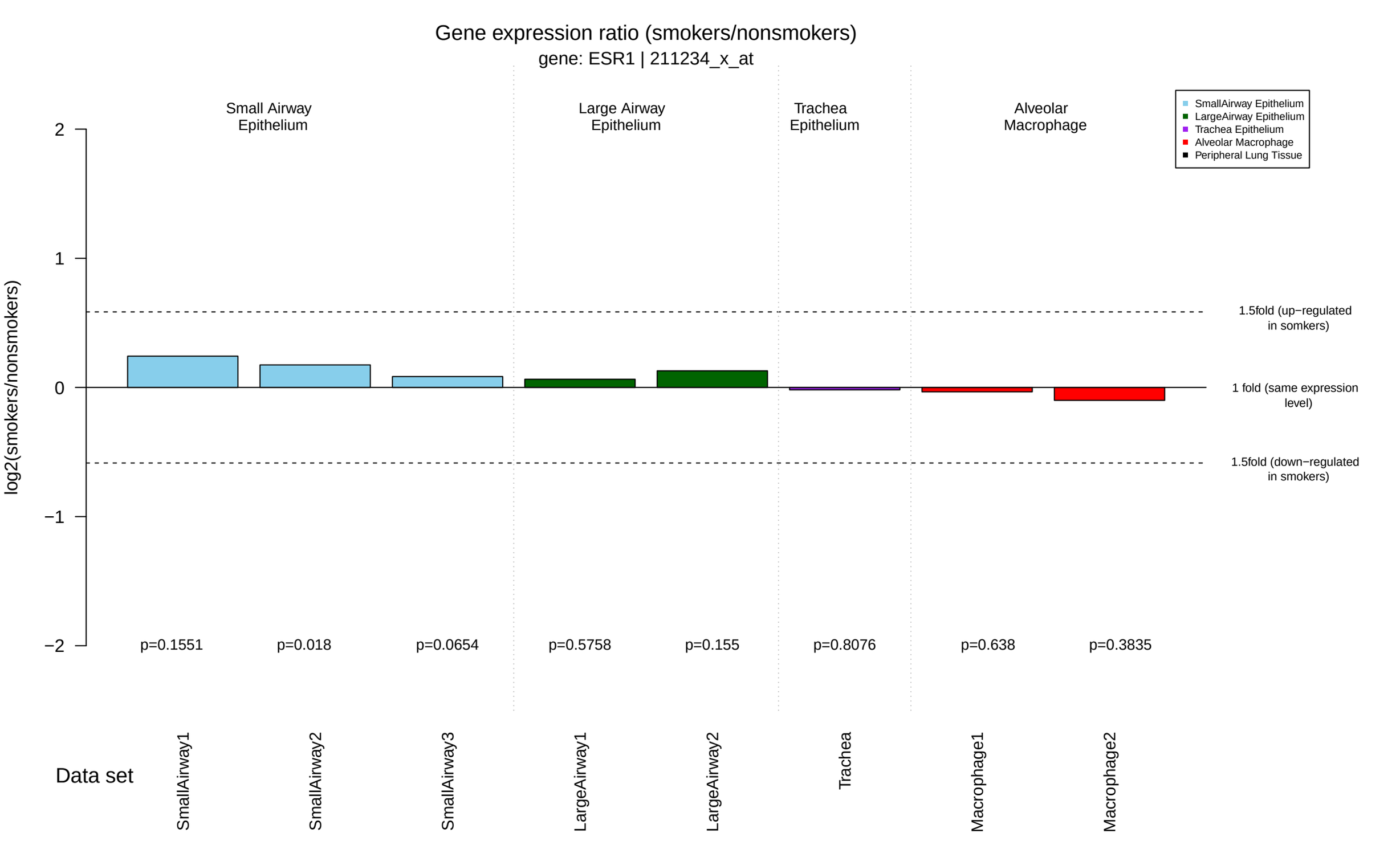
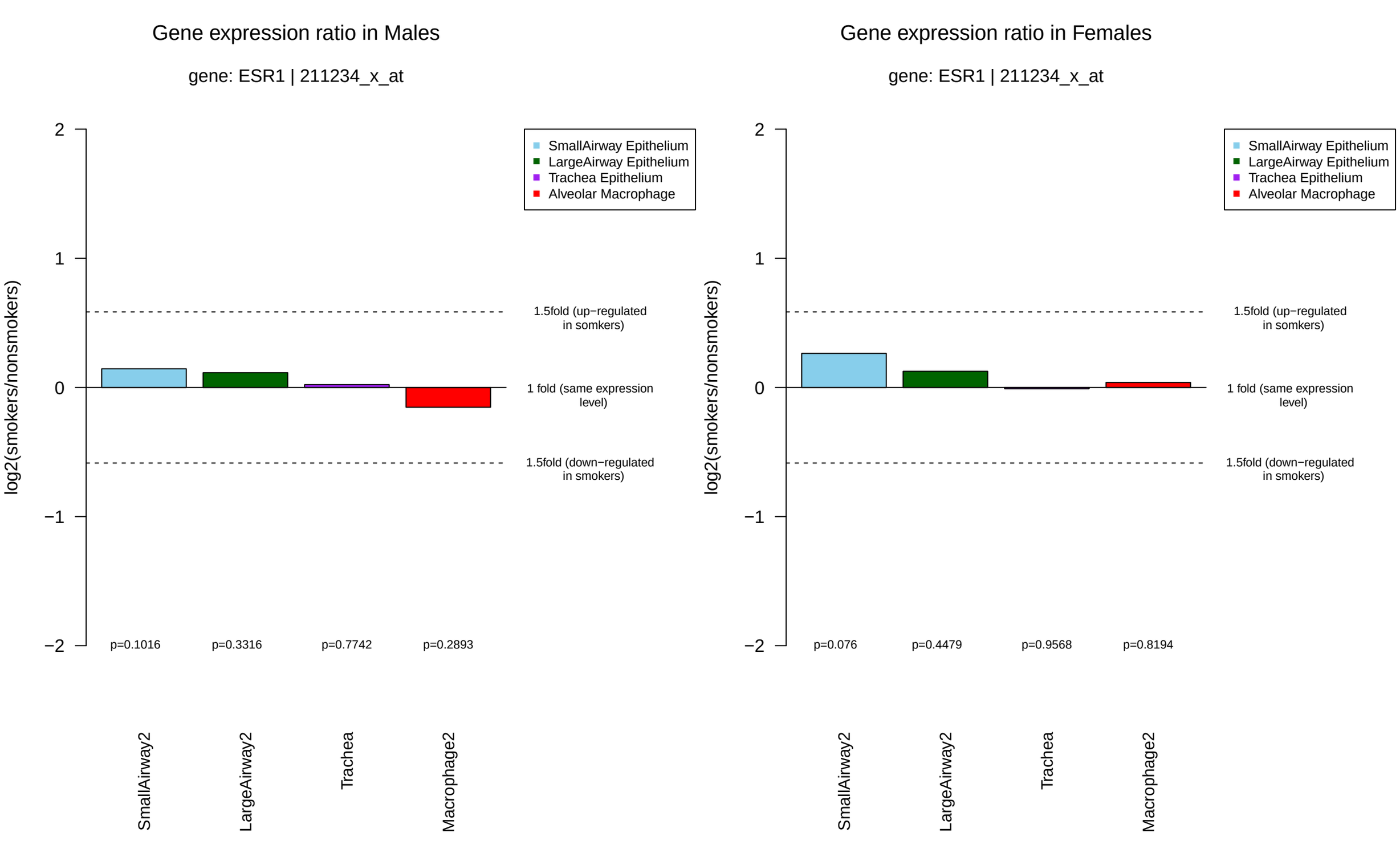
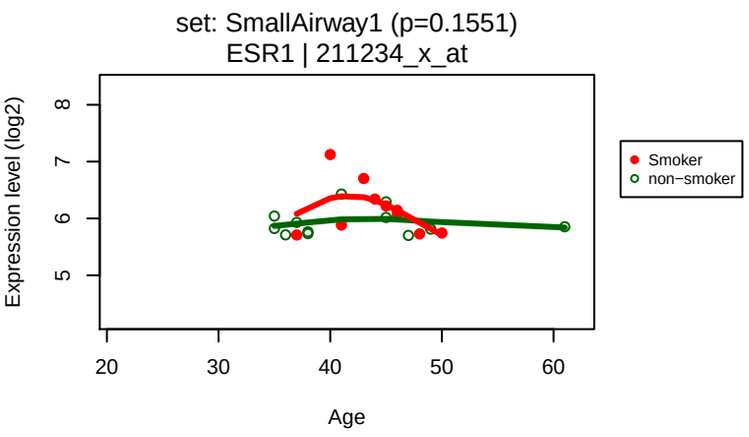
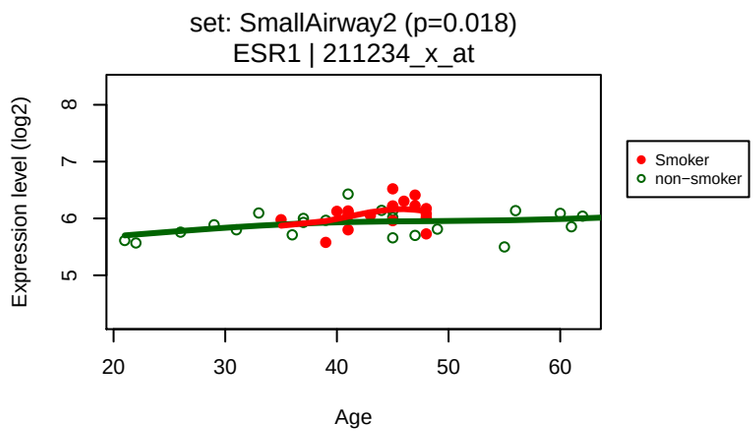
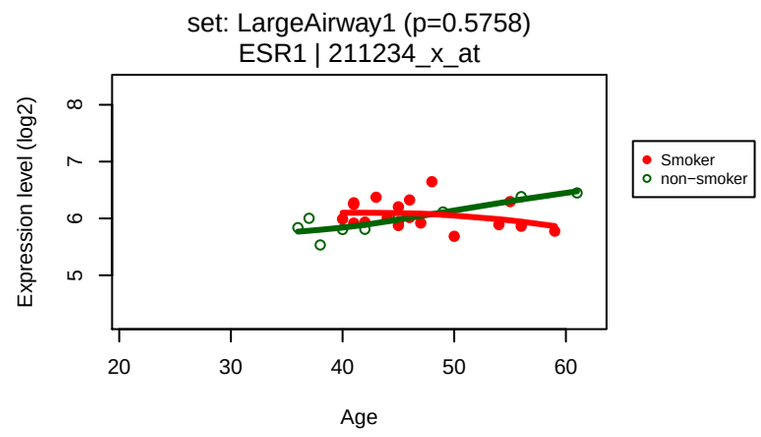

Gene expression of smokers and nonsmokers
gene: ESR1 | 211234_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1551
p=0.018
p=0.0654
p=0.5758
p=0.155
p=0.8076
p=0.638
p=0.3835
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

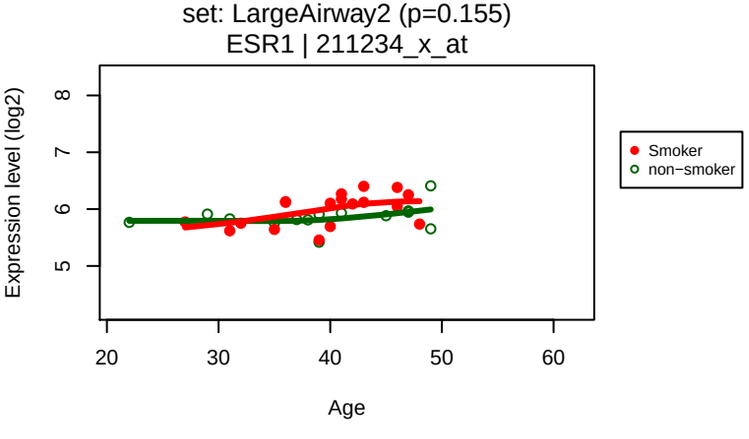

Gene expression ratio (smokers/nonsmokers)
gene: ESR1 | 211234_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1551
p=0.018
p=0.0654
p=0.5758
p=0.155
p=0.8076
p=0.638
p=0.3835
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

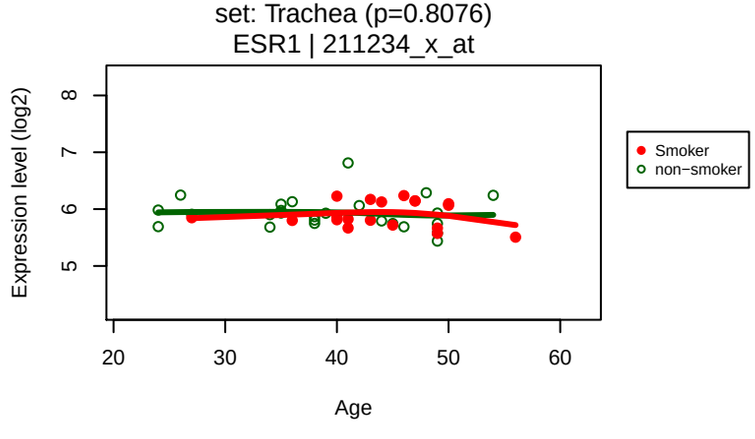

Gene expression ratio in Males
gene: ESR1 | 211234_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1016
p=0.3316
p=0.7742
p=0.2893
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ESR1 | 211234_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.076
p=0.4479
p=0.9568
p=0.8194
SmallAirway2
LargeAirway2
Trachea
Macrophage2

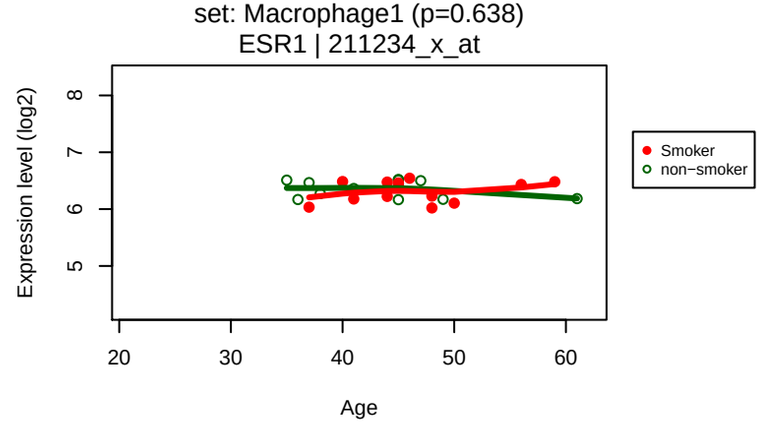

set: SmallAirway1 (p=0.1551)
ESR1 | 211234_x_at
Expression level (log2)
Age
Smoker
non−smoker

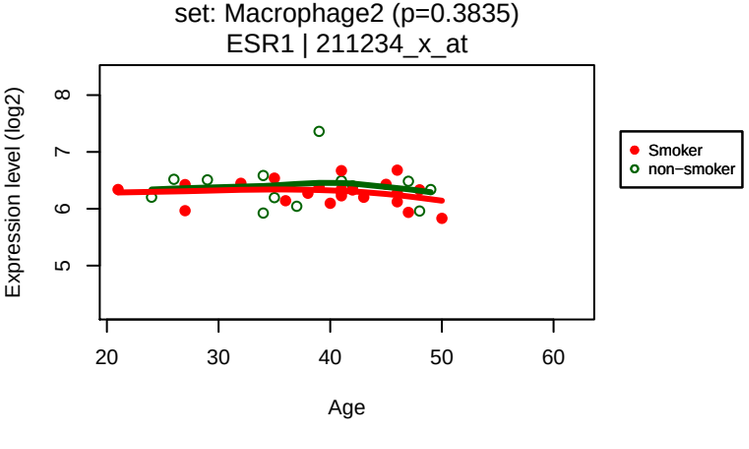

set: SmallAirway2 (p=0.018)
ESR1 | 211234_x_at
Expression level (log2)
Age
Smoker
non−smoker

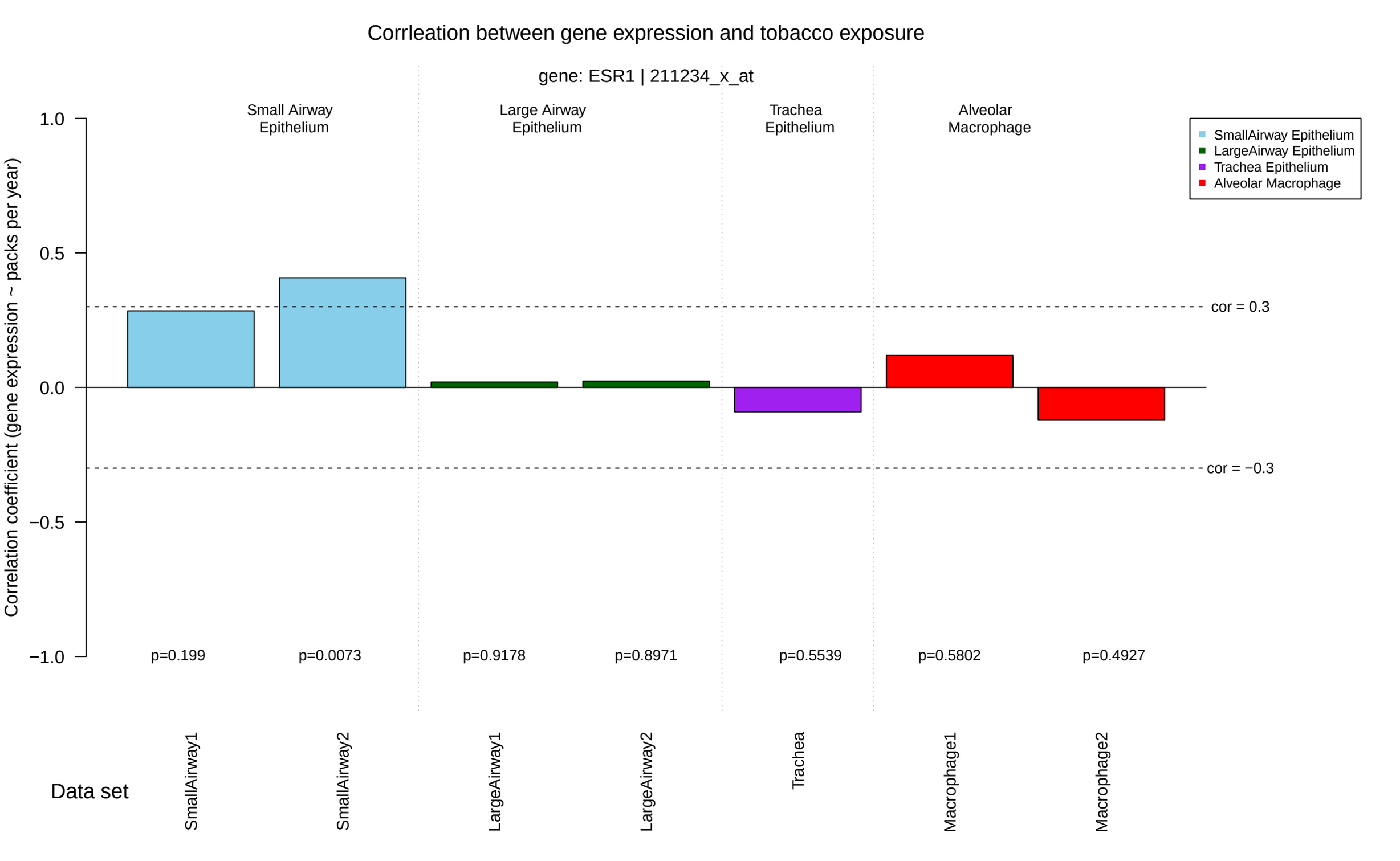

set: LargeAirway1 (p=0.5758)
ESR1 | 211234_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.155)
ESR1 | 211234_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8076)
ESR1 | 211234_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.638)
ESR1 | 211234_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3835)
ESR1 | 211234_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ESR1 | 211234_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.199
p=0.0073
p=0.9178
p=0.8971
p=0.5539
p=0.5802
p=0.4927
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2